

Integration of Onboard Charger and Wireless Charging System For Electric Vehicles With Shared Coupler, Compensation, and Rectifier

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Abstract—For electric vehicles (EVs) equipped with an onboard charger (OBC) and a wireless power transfer (WPT) system, there will be two charging systems in the EVs, which increases the cost, weight, and complexity. Based on the proposed design of magnetic couplers, this letter proposes a novel integration method for OBC and WPT systems of EV charging. The isolation transformer of the OBC system can be regarded as two strongly coupled coils. The secondary-side coil, namely the receiving coil, can also be loosely coupled with the transmitting coil of the WPT system. Thus, the receiving coil, the compensation, and the rectifier of both the OBC and WPT systems can be shared. In this way, the EV charging system can be capable of conductive charging and wireless charging while still having the advantages of cost effectiveness and high power density. An experimental prototype is implemented to validate the proposal.

Index Terms—Integration, interoperability, onboard charger (OBC), shared, wireless power transfer (WPT).

I. INTRODUCTION

ELECTRIC vehicles (EVs) have been developing rapidly in the recent decade [1]. EVs have advantages, such as energy efficiency, emission reduction, high performance, and low maintenance. However, there are also disadvantages, such as high cost, range anxiety, and the charging issues. The limited number of charging stations and the long charging time are the major problems of EV charging. In addition, the EV charging technology can also be optimized.

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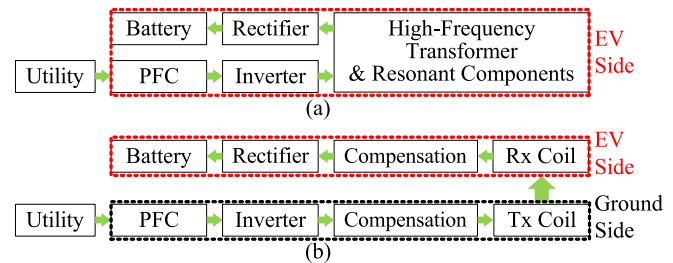


Fig. 1. Block diagram for EV charging. (a) OBC. (b) WPT.

There are two methods for EV charging: conductive charging and wireless charging. Conductive charging would be efficient, cost effective, and of good electromagnetic compatibility. Presently, it is the mainstream charging technique for EVs. In comparison, wireless charging has the advantages of automation, convenience, and of weatherproof ability [2], [3], [4]. It has attracted much attention in recent years [5], [6], [7], [8], [9].

For an EV conductive charging system with a utility ac input, it can be referred as the onboard charger (OBC), whose topology is shown in Fig. 1(a). Power factor correction (PFC) transforms the ac power from the utility to dc. An isolated dc–dc converter, consisting of an inverter, a high-frequency transformer (namely two strongly coupled coils) with resonant components, such as capacitors and inductors, and a rectifier, is added for level conversion and isolation purposes. Finally, the dc power feeds the battery. For a wireless power transfer (WPT) system for EVs, its topology is given in Fig. 1(b). Similar power conversion stages are observed in the WPT system except that the transmitting (Tx) coil and the receiving (Rx) coil constitute loosely coupled coils.

For EVs with both conductive and wireless charging capabilities, there would be two systems in the EV, namely the OBC system and the WPT system. This would increase the cost, weight, and complexity, and decrease the power density of the EV charging system, which is crucial since it is installed in the EV. As can be seen from Fig. 1, there are some similar power conversion stages for the OBC and WPT systems, such as the coupled coils and the rectifier. Some integration techniques have been utilized. The rectifier was shared in [10] for both wired and wireless charging. The Rx coil for the WPT system was added to the existing OBC system [11] with a *CLLC* converter to share the rectifier and part of the resonant capacitors. The dc–dc inductor on the Rx side was integrated with the Rx coil [12]. For these

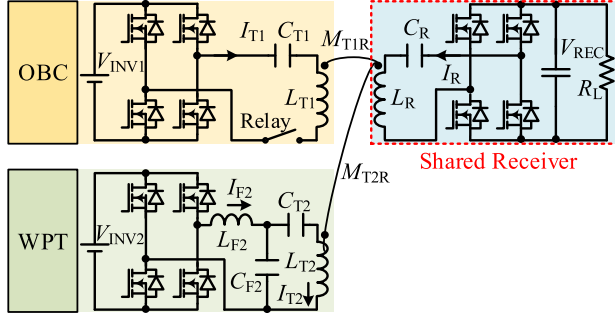


Fig. 2. Proposed integration topology for OBC and WPT systems.

existing methods, the couplers for the WPT and OBC systems are separate and independent. The integration and interoperability of the WPT and OBC systems can be further improved.

This letter proposes a novel integration method for the WPT and OBC systems with a shared receiver-side coupler, compensation, and rectifier. In this way, the EV charging system can be cost effective, lightweight, and of high power density.

II. PROPOSED INTEGRATION METHOD

A. Topology

The proposed integration topology for the OBC and WPT systems is shown in Fig. 2. The PFC is not shown in the OBC system. L_{T1} and L_{T2} are the respective Tx coils of the OBC and WPT systems. L_R is the Rx coil. L_{F2} is the compensating inductor. C_{T1} , C_{T2} , C_R , and C_{F2} are their resonant capacitors. V_{INV1} and V_{INV2} are the inverter dc voltages. V_{REC} is the rectifier dc voltage. I_{T1} , I_{T2} , I_R , and I_{F2} are their currents. M_{T1R} and M_{T2R} are the mutual inductances. R_L is the load resistance.

L_{T1} and L_R are two strongly coupled coils, constituting the transformer. With C_{T1} and C_R , a CLLC converter is formed. For the WPT system, the inductor–capacitor–capacitor–series (LCC-S) topology is utilized so that the output power decreases with the increasing misalignment to enable misalignment tolerance. Note that the primary-side compensation of the WPT system is not necessarily LCC and can be other compensations such as series, forming an S-S topology.

The system works at the resonant frequency ω_0 , which is as follows:

$$\omega_0 = \frac{1}{\sqrt{L_R C_R}} = \frac{1}{\sqrt{L_{T1} C_{T1}}} = \frac{1}{\sqrt{L_{F2} C_{F2}}} = \frac{1}{\sqrt{L_{T2} \frac{C_{T2} C_{F2}}{C_{T2} + C_{F2}}}}. \quad (1)$$

B. WPT Mode

When the charging system works in the WPT mode, the topology and the equivalent circuit are shown in Fig. 3. The relay is open to avoid the impact on the OBC transmitter. R_{T2} , R_{F2} , and R_R are the equivalent resistances of the Tx coil, the compensating inductor, and the Rx coil of the WPT system, respectively. U_{T2} is the fundamental component of the inverter ac voltage. R_{EQ} is the equivalent ac load resistance. Based on fundamental harmonic approximation (FHA), they can be

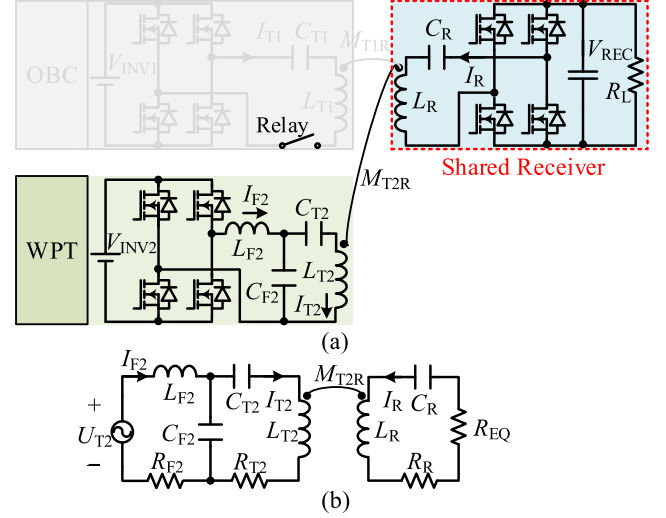


Fig. 3. WPT mode. (a) Topology. (b) Equivalent circuit.

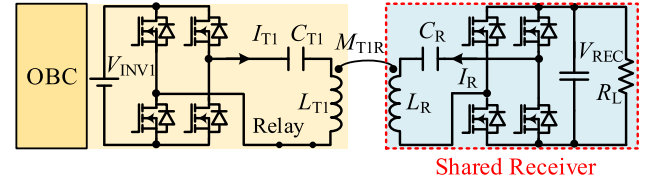


Fig. 4. OBC mode: topology.

expressed as

$$U_{T2} = \frac{2\sqrt{2}}{\pi} V_{INV2}, R_{EQ} = \frac{8}{\pi^2} R_L. \quad (2)$$

At ω_0 , ignoring R_{F2} , R_{T2} , and R_R yields

$$I_{F2} = \left(\frac{M_{T2R}}{L_{F2}} \right)^2 \frac{U_{T2}}{R_{EQ}}, I_{T2} = \frac{U_{T2}}{\omega_0 L_{F2}}, I_R = \frac{M_{T2R}}{L_{F2}} \frac{U_{T2}}{R_{EQ}}. \quad (3)$$

The output power can be calculated as follows:

$$P_{OUT} = \left(\frac{M_{T2R}}{L_{F2}} \right)^2 \frac{U_{T2}^2}{R_{EQ}}. \quad (4)$$

The ac–ac efficiency can be expressed as follows:

$$\eta = \frac{P_{OUT}}{P_{OUT} + I_R^2 R_R + I_{T2}^2 R_{T2} + I_{F2}^2 R_{F2}} = \frac{1}{1 + \left[\left(\frac{M_{T2R}}{L_{F2}} \right)^2 R_{F2} + R_R \right] \frac{1}{R_{EQ}} + \frac{R_{T2} R_{EQ}}{(\omega_0 M_{T2R})^2}}. \quad (5)$$

C. OBC Mode

When the charging system works in the OBC mode, the topology is shown in Fig. 4. The WPT transmitter is not present. Due to the strong coupling of the coils L_{T1} and L_R , FHA is no longer effective [13]. Time-domain analysis can be carried on, similar to [14]. Since it is not the focus of this letter, the modeling process is not presented here. Circuit simulation can

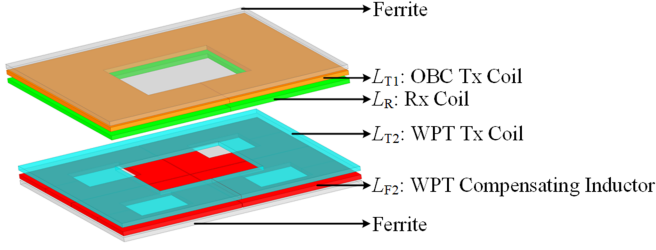


Fig. 5. One possible solution for magnetic couplers of proposed method.

TABLE I
SIMULATED AND MEASURED COIL PARAMETERS

Value	L_{T1} (μH)	L_{T2} (μH)	L_R (μH)	L_{F2} (μH)	k_{T1R}
Sim.	191.1	166.8	167.2	109.1	0.901
Meas.	189.9	161.7	166.0	107.7	0.866
Value	k_{T2R}	k_{T12}	k_{F2T1}	k_{F2T2}	k_{F2R}
Sim.	0.292	0.265	0.0009	0.0045	0.0015
Meas.	0.276	0.254	0.0081	0.0270	0.0078

also be carried out to predict the performance of the OBC system. In this mode, the self-inductances are compensated by the two capacitors C_{T1} and C_R .

D. Magnetic Coupler Design

The Rx coil should be strongly coupled with the Tx coil of the OBC system and loosely coupled with the Tx coil of the WPT system. Thus, the magnetic flux distribution of the transformer should not be the same as the traditional one. Instead, it should be able to link with another coil placed in a certain distant way. The compensating inductor L_{F2} of the WPT system can also be integrated with the existing couplers. To achieve decoupling on two perpendicular directions, L_{F2} is designed as a quadrupole coil. In this way, L_{F2} is decoupled with all the other coils in two perpendicular directions. The proposed solution for the magnetic couplers is shown in Fig. 5. The Rx coil is stacked with the Tx coil of the OBC system. The compensating inductor is also stacked with the Tx coil of the WPT system. The ferrite is only on one side. In this way, the Rx coil achieves strong coupling with the Tx coil of the OBC system and weak coupling with the Tx coil of the WPT system.

For the prototype, the sizes of all the coils are set to 300×300 mm. The winding widths of the unipolar coils are set to 85 mm. The air gap between the Tx coil of the WPT system and the Rx coil is 100 mm. Under such a setup, the simulated and measured self-inductances and coupling coefficients are given in Table I, where k_{T1R} , k_{T2R} , k_{T12} , k_{F2T1} , k_{F2T2} , and k_{F2R} are the coupling coefficients of the respective coils. L_{F2} achieves almost zero coupling with the other coils. The simulated results match well with the measured ones. k_{T1R} is large enough so that the transformer with strong coupling can be implemented. k_{T2R} is also acceptable for WPT systems. The proposed design of magnetic couplers is expected to work for the proposed system.

Compared with the conventional solution containing two sets of receivers (receiving coil, compensation network, and rectifier) for the OBC and WPT systems, respectively, the proposed

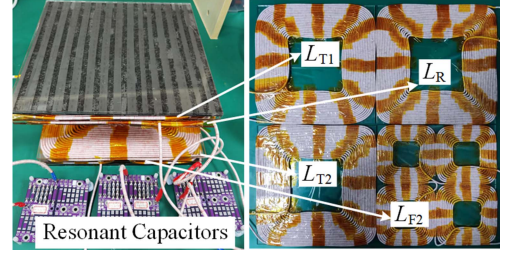


Fig. 6. Photo of experimental prototype.

TABLE II
PARAMETERS OF EXPERIMENTAL PROTOTYPE

V_{INV}	f	C_R	C_{T1}	C_{T2}	C_{F2}
350 V	85 kHz	21.1 nF	18.5 nF	64.9 nF	32.6 nF

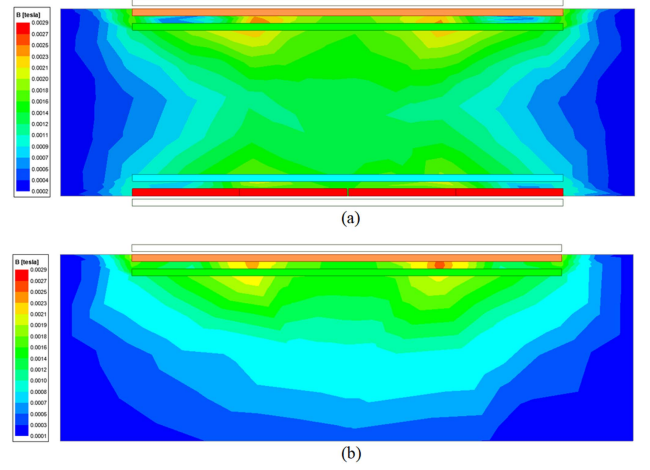


Fig. 7. Simulated magnetic field distribution. (a) WPT. (b) OBC.

method would be beneficial in terms of volume, weight, and cost. However, in terms of efficiency, for the OBC mode, the leakage inductance for the OBC transformer would be larger than the conventional one, leading to extra losses; for the WPT mode, the distance between the WPT Tx coil and the Rx coil, or the distance between the Rx coil and the Rx ferrite, would increase, leading to a slight drop of the coupling coefficient. Considering these two issues, the efficiency of both modes would be slightly affected.

III. SIMULATION AND EXPERIMENTAL VALIDATION

A prototype is implemented to validate the proposal, which is shown in Fig. 6. The parameters are given in Table II. The simulated magnetic field distributions of the two modes are shown in Fig. 7.

R_L changes from 20 to 60 Ω . The output power and the dc-dc efficiency of the WPT and OBC systems are depicted in Fig. 8, respectively. It can be seen that relatively high efficiency can be achieved for both systems. The functions of both the WPT and OBC systems have been realized. The experimental waveforms are shown in Fig. 9.

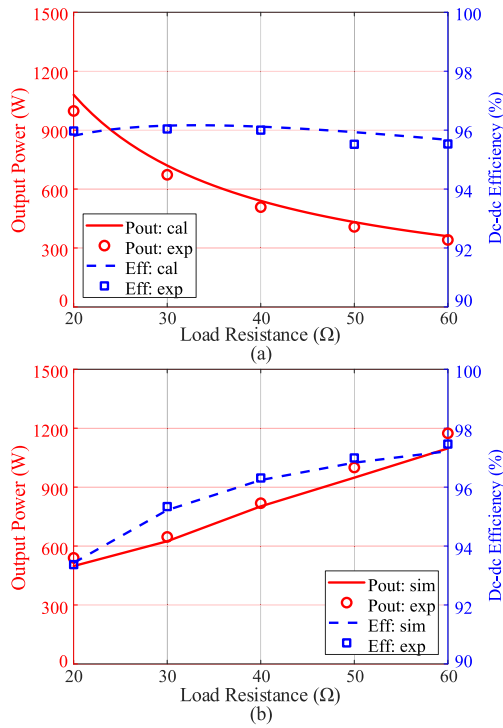


Fig. 8. Calculated, simulated and measured output power and dc-dc efficiency. (a) WPT. (b) OBC.

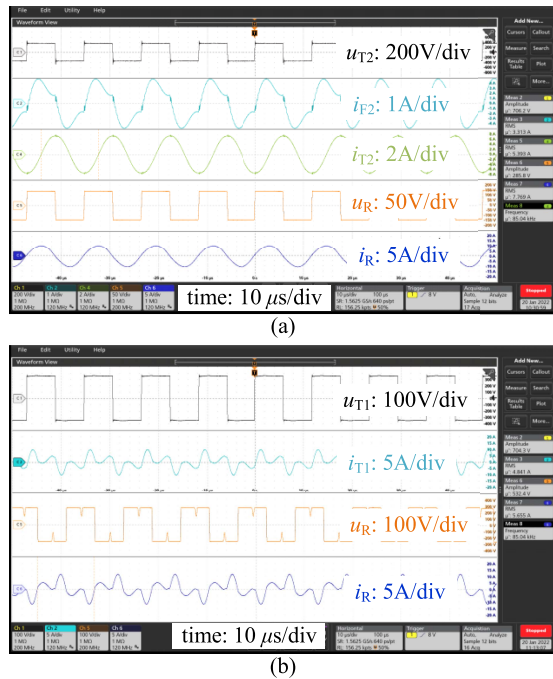


Fig. 9. Experimental waveforms. (a) WPT. (b) OBC.

For the variation of the output power with the load resistance, the OBC system behaves oppositely compared with the WPT system. This is caused by the different transfer characteristics. Please note that another power conversion stage, such as a dc-dc converter, can be added for regulation purposes.

IV. CONCLUSION

A novel integration method for the WPT and OBC systems for EV charging was proposed in this letter. By specifically designing the strongly coupled transformer, the secondary-side inductor, namely the Rx coil, was coupled with the Tx coil of the WPT system. Thus, the Rx coil, the compensation, and the rectifier were shared for both the WPT and OBC systems. The charging performances of the WPT and OBC systems were evaluated. In this way, the EV charging system was cost effective, lightweight, and of high power density. The experimental results validated the effectiveness of the proposal.

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